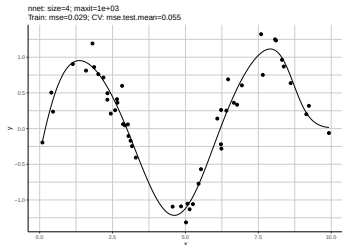
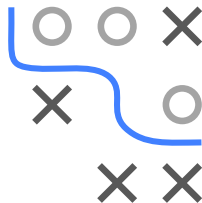


Introduction to Machine Learning

Neural Networks

Universal Approximation



Learning goals

- Universal approximation theorem for one-hidden-layer neural networks
- The pros and cons of a low approximation error

UNIVERSAL APPROXIMATION PROPERTY I

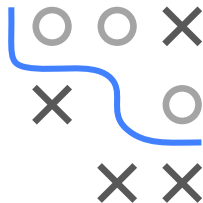
Theorem. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, non-constant, bounded, and monotonically increasing function. Let $C \subset \mathbb{R}^p$ be compact, and let $\mathcal{C}(C)$ denote the space of continuous functions $C \rightarrow \mathbb{R}$. Then, given a function $g \in \mathcal{C}(C)$ and an accuracy $\varepsilon > 0$, there exists a hidden layer size $m \in \mathbb{N}$ and a set of coefficients $W_j \in \mathbb{R}^p$, $u_j, b_j \in \mathbb{R}$ (for $j \in \{1, \dots, m\}$), such that

$$f : C \rightarrow \mathbb{R}; \quad f(\mathbf{x}) = \sum_{j=1}^m u_j \cdot \sigma(W_j^T \mathbf{x} + b_j)$$

is an ε -approximation of g , that is,

$$\|f - g\|_{\infty} := \max_{\mathbf{x} \in C} |f(\mathbf{x}) - g(\mathbf{x})| < \varepsilon .$$

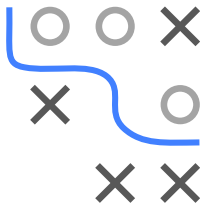
The theorem extends trivially to multiple outputs.



UNIVERSAL APPROXIMATION PROPERTY II

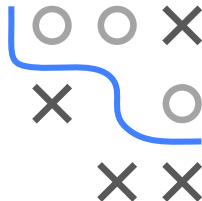
Corollary. Neural networks with a single sigmoidal hidden layer and linear output layer are universal approximators.

- This means that for a given target function g there exists a sequence of networks $(f_k)_{k \in \mathbb{N}}$ that converges (pointwise) to g .
- Usually, as the networks come closer and closer to g , they will need more and more hidden neurons.
- A network with fixed layer sizes can only model a subspace of all continuous functions.
- The continuous functions form an infinite dimensional vector space. Therefore arbitrarily large hidden layer sizes are needed.



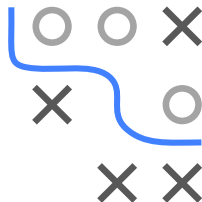
UNIVERSAL APPROXIMATION PROPERTY III

- Why is universal approximation a desirable property?
- Recall the definition of a Bayes optimal hypothesis $f^* : X \rightarrow Y$. It is the best possible hypothesis (model) for the given problem: it has minimal loss averaged over the data generating distribution.
- So ideally we would like the neural network (or any other learner) to approximate the Bayes optimal hypothesis.
- Usually we do not manage to learn f^* .
- This is because we do not have enough (infinite) data. We have no control over this, so we have to live with this limitation.
- But we do have control over which model class we use.



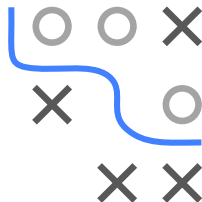
UNIVERSAL APPROXIMATION PROPERTY IV

- Universal approximation $\Rightarrow$ approximation error tends to zero as hidden layer size tends to infinity.
- Positive approximation error implies that no matter how big the data set, we cannot find the optimal model.
- This bears the risk of systematic under-fitting, which can be avoided with a universal model class.



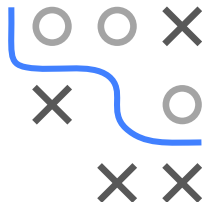
UNIVERSAL APPROXIMATION PROPERTY V

- As we know, there are also good reasons for restricting the model class.
- This is because a flexible model class with universal approximation ability often results in over-fitting, which is no better than under-fitting.
- Thus, “universal approximation $\Rightarrow$ low approximation error”, but at the risk of a substantial learning error.
- In general, models of intermediate flexibility give the best predictions. For neural networks this amounts to a reasonably sized hidden layer.



EXAMPLE : REGRESSION/CLASSIFICATION

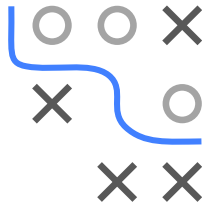
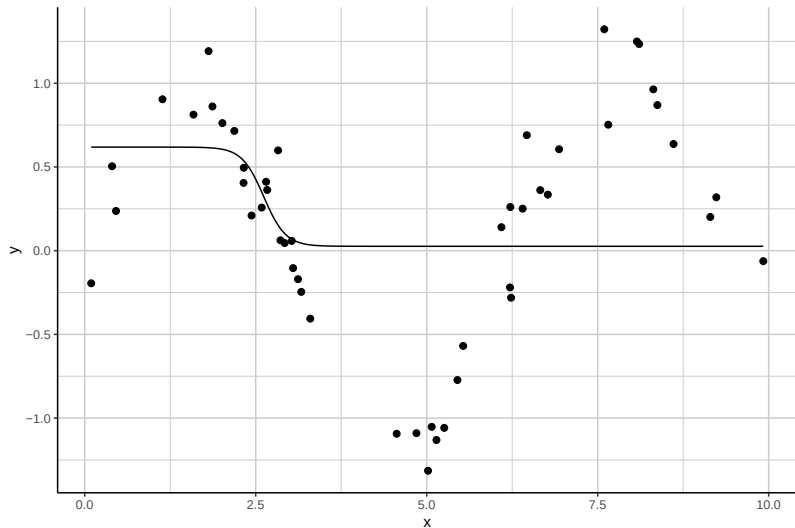
- Let's look at a few examples of the types of functions and decisions boundaries learnt by neural networks (with a **single** hidden layer) of various sizes.
- "size" here refers to the number of neurons in the hidden layer.
- The number of "iterations" in the following slides corresponds to the number of steps of the applied iterative optimization algorithm (stochastic gradient descent).



REGRESSION EX.: 1000 TRAINING ITERATIONS I

nnet: size=1; maxit=1e+03

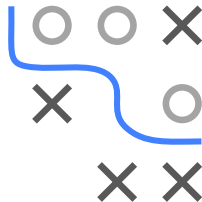
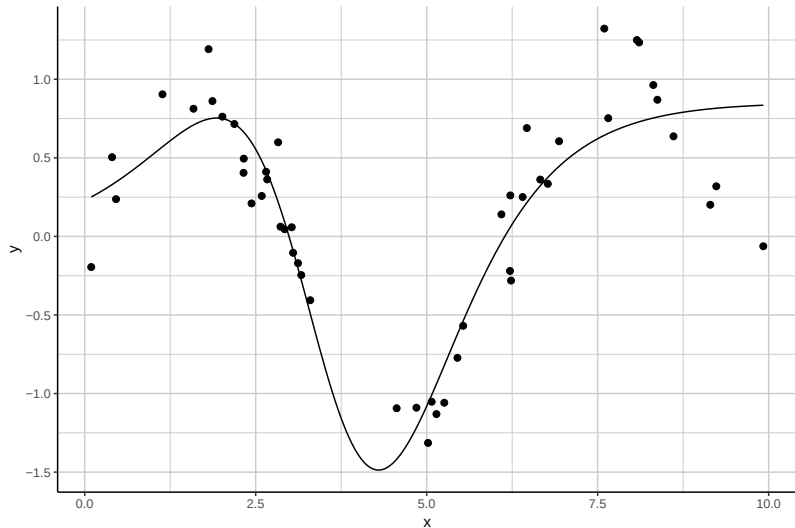
Train: mse=0.391; CV: mse.test.mean=0.419



REGRESSION EX.: 1000 TRAINING ITERATIONS II

nnet: size=2; maxit=1e+03

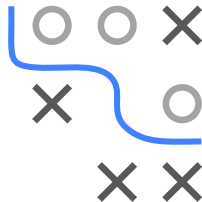
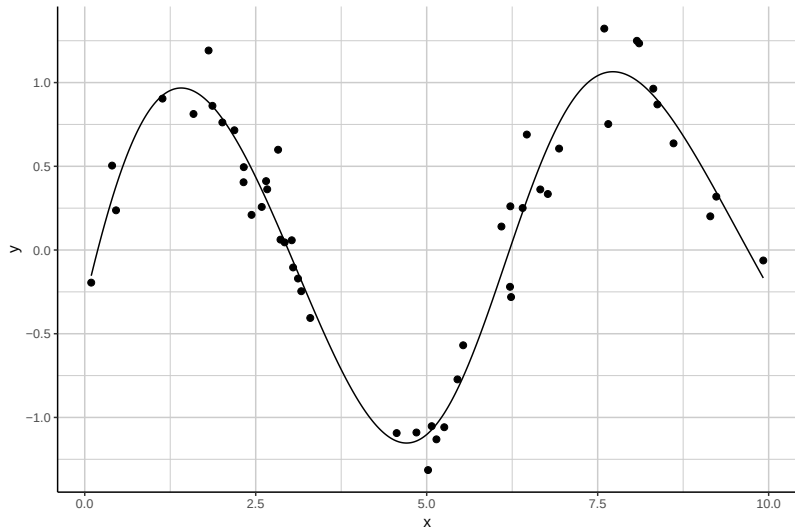
Train: mse=0.088; CV: mse.test.mean=0.112



REGRESSION EX.: 1000 TRAINING ITERATIONS III

nnet: size=3; maxit=1e+03

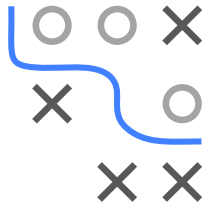
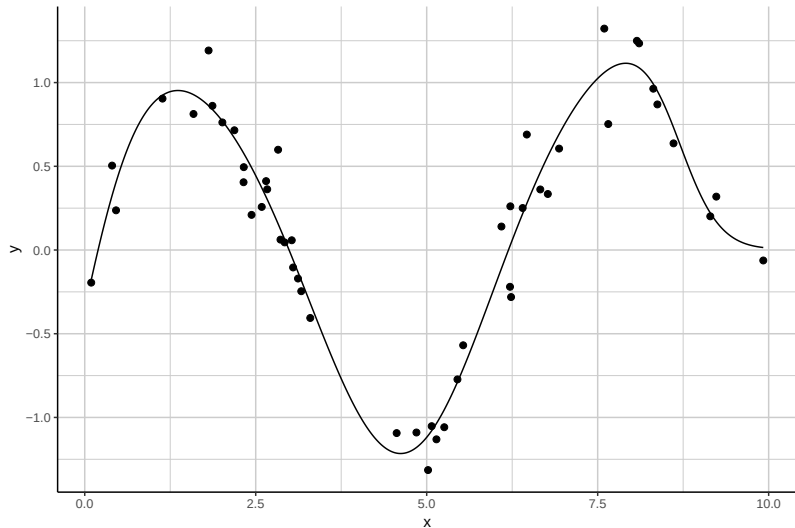
Train: mse=0.032; CV: mse.test.mean=0.063



REGRESSION EX.: 1000 TRAINING ITERATIONS IV

nnet: size=4; maxit=1e+03

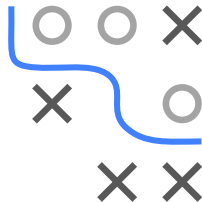
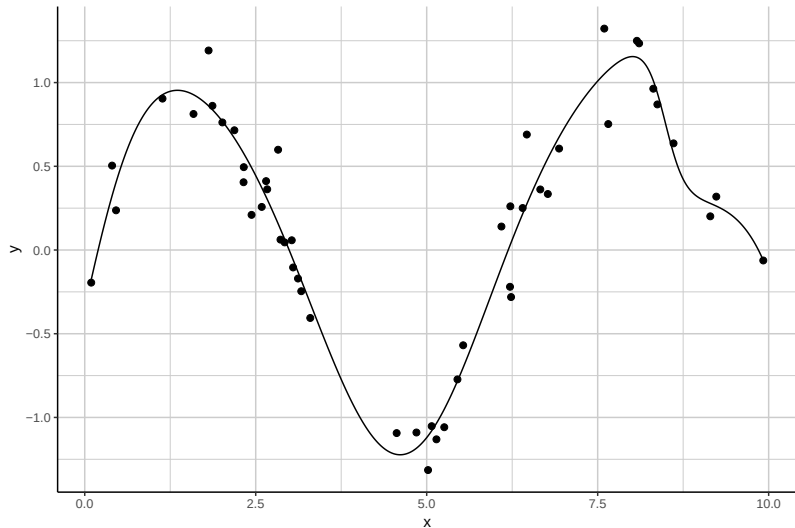
Train: mse=0.029; CV: mse.test.mean=0.055



REGRESSION EX.: 1000 TRAINING ITERATIONS V

nnet: size=5; maxit=1e+03

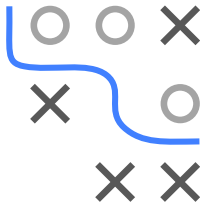
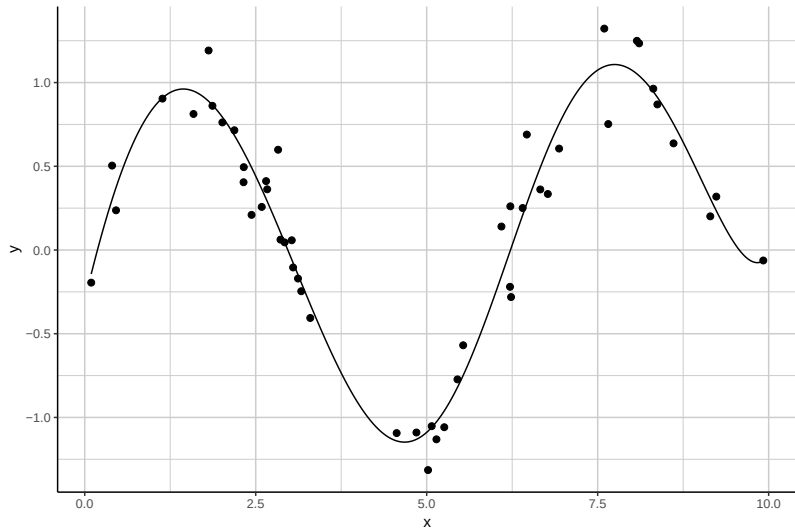
Train: mse=0.028; CV: mse.test.mean=19.845



REGRESSION EX.: 1000 TRAINING ITERATIONS VI

nnet: size=6; maxit=1e+03

Train: mse=0.031; CV: mse.test.mean=4.374

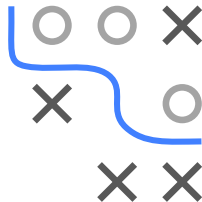
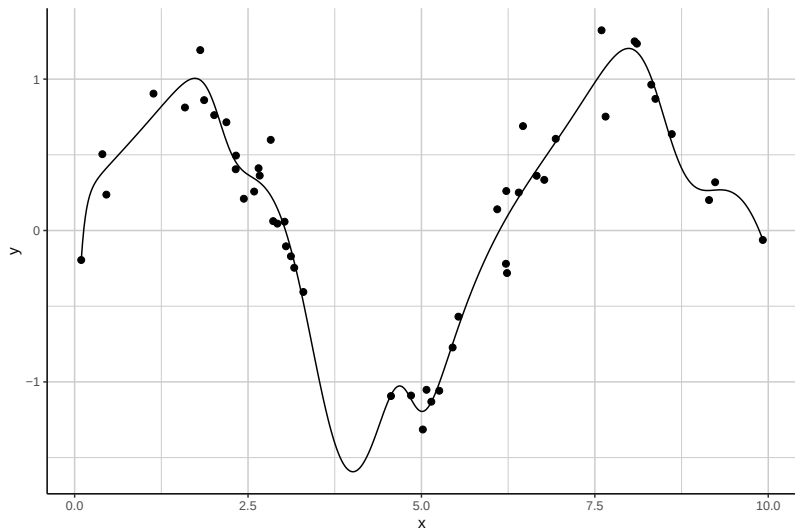


REGRESSION EX.: 1000 TRAINING ITERATIONS

VII

nnet: size=10; maxit=1e+03

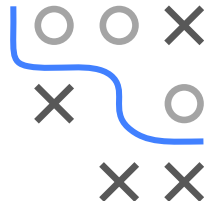
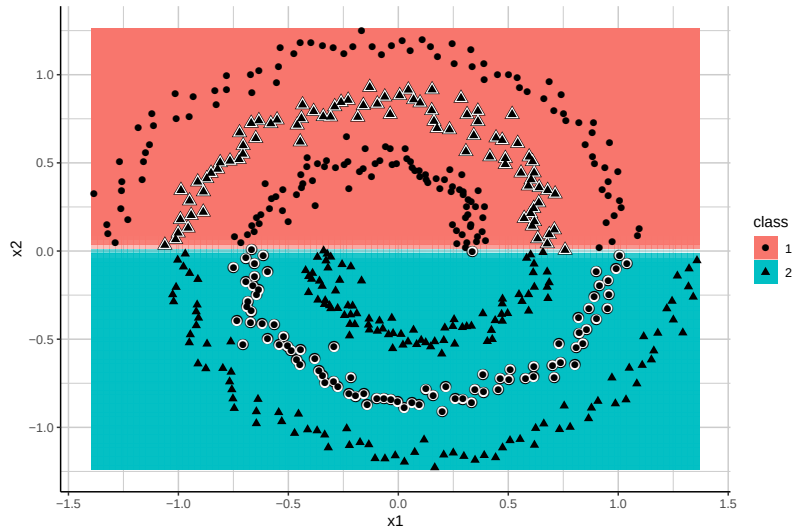
Train: mse=0.023; CV: mse.test.mean=0.698



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=1; maxit=500

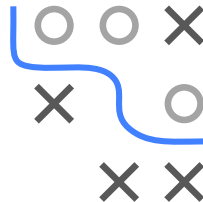
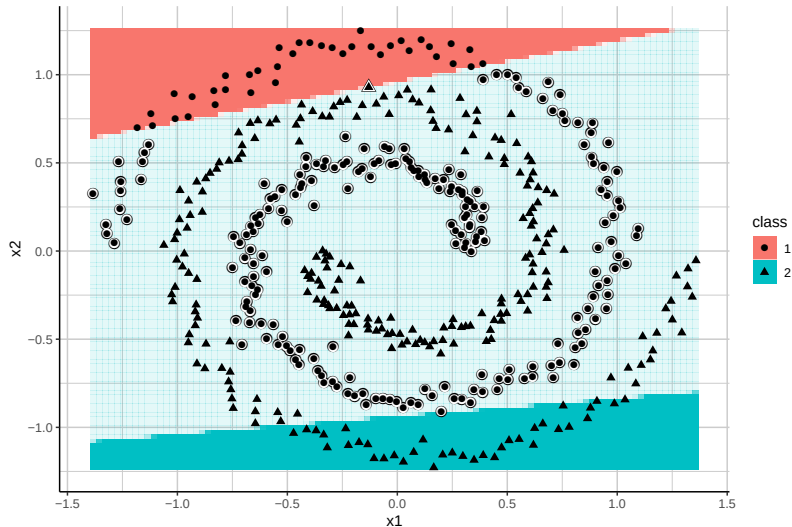
Train: mmce=0.336; CV: mmce.test.mean=0.346



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=2; maxit=500

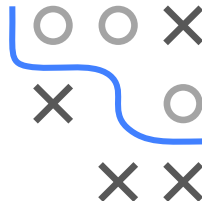
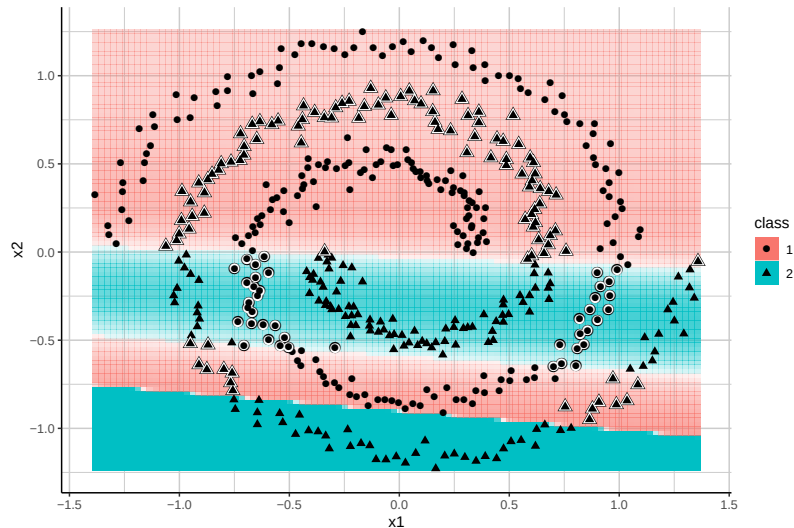
Train: mmce=0.426; CV: mmce.test.mean=0.412



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=3; maxit=500

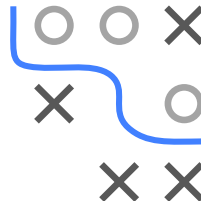
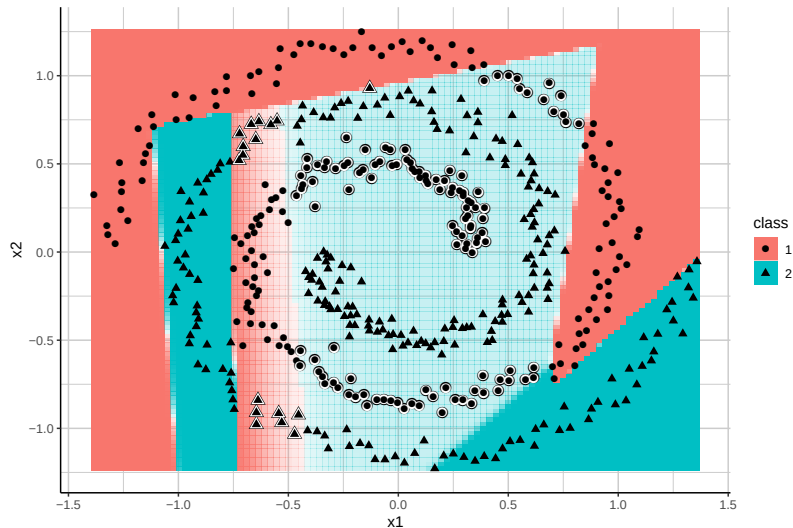
Train: mmce=0.290; CV: mmce.test.mean=0.374



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=5; maxit=500

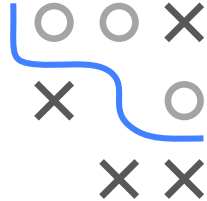
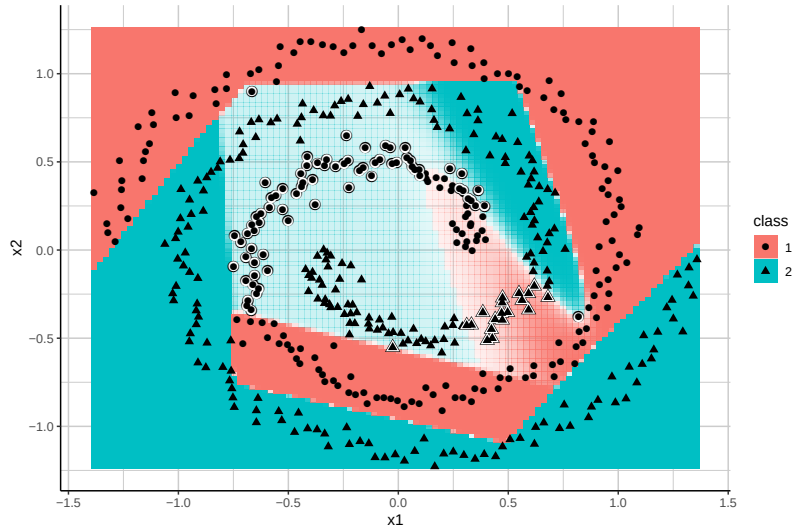
Train: mmce=0.272; CV: mmce.test.mean=0.322



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=10; maxit=500

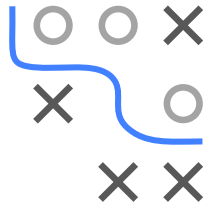
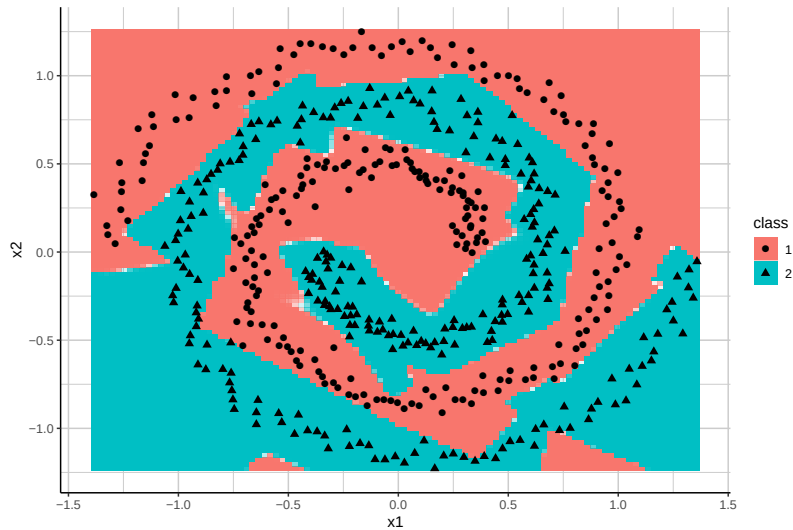
Train: mmce=0.184; CV: mmce.test.mean=0.106



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=30; maxit=500

Train: mmce=0.000; CV: mmce.test.mean=0.034



CLASSIFICATION: 500 TRAINING ITERATIONS

nnet: size=50; maxit=500

Train: mmce=0.000; CV: mmce.test.mean=0.026

